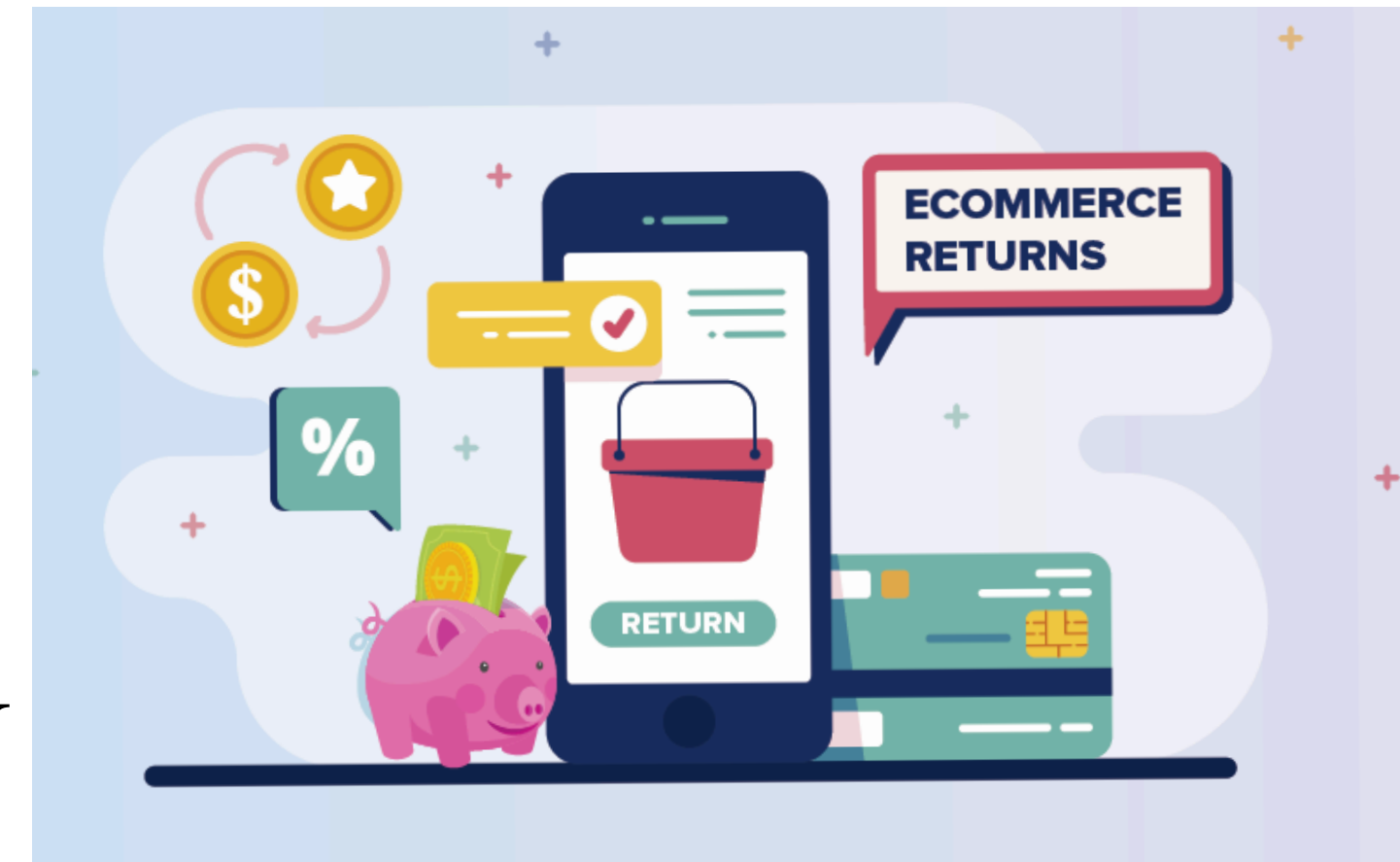


# E-COMMERCE PRODUCT RETURN ANALYTICS AND CUSTOMERS SATISFACTION



Presented By:

**V.HARIPRIYA**

# PROBLEM STATEMENT

**E-commerce businesses face high product return rates, leading to significant revenue loss and reduced customer satisfaction.**

**However, the key factors driving returns are not clearly understood.**

**This project aims to analyze return patterns and identify the main causes to reduce returns and improve business performance.**



# OBJECTIVE

-  **To analyze product return patterns in e-commerce .**
-  **To identify key factors influencing returns (category, delivery time, discount, rating) .**
-  **To measure the financial impact of returns on business .**
-  **To provide data-driven recommendations to reduce return rates .**



# DATASET

- **Synthetic E-commerce dataset .**
- **Contains approximately 100,000 records .**
- **Includes 17+ columns .**
- **Key fields: Product Category, Region, Discount %, Delivery Days, Customer Rating, Return Status .**
- **Dataset represents customer orders and return behavior .**



# TOOLS USED

- **Microsoft Excel – Data cleaning and preprocessing .**
- **Power BI – Data visualization and dashboard creation .**



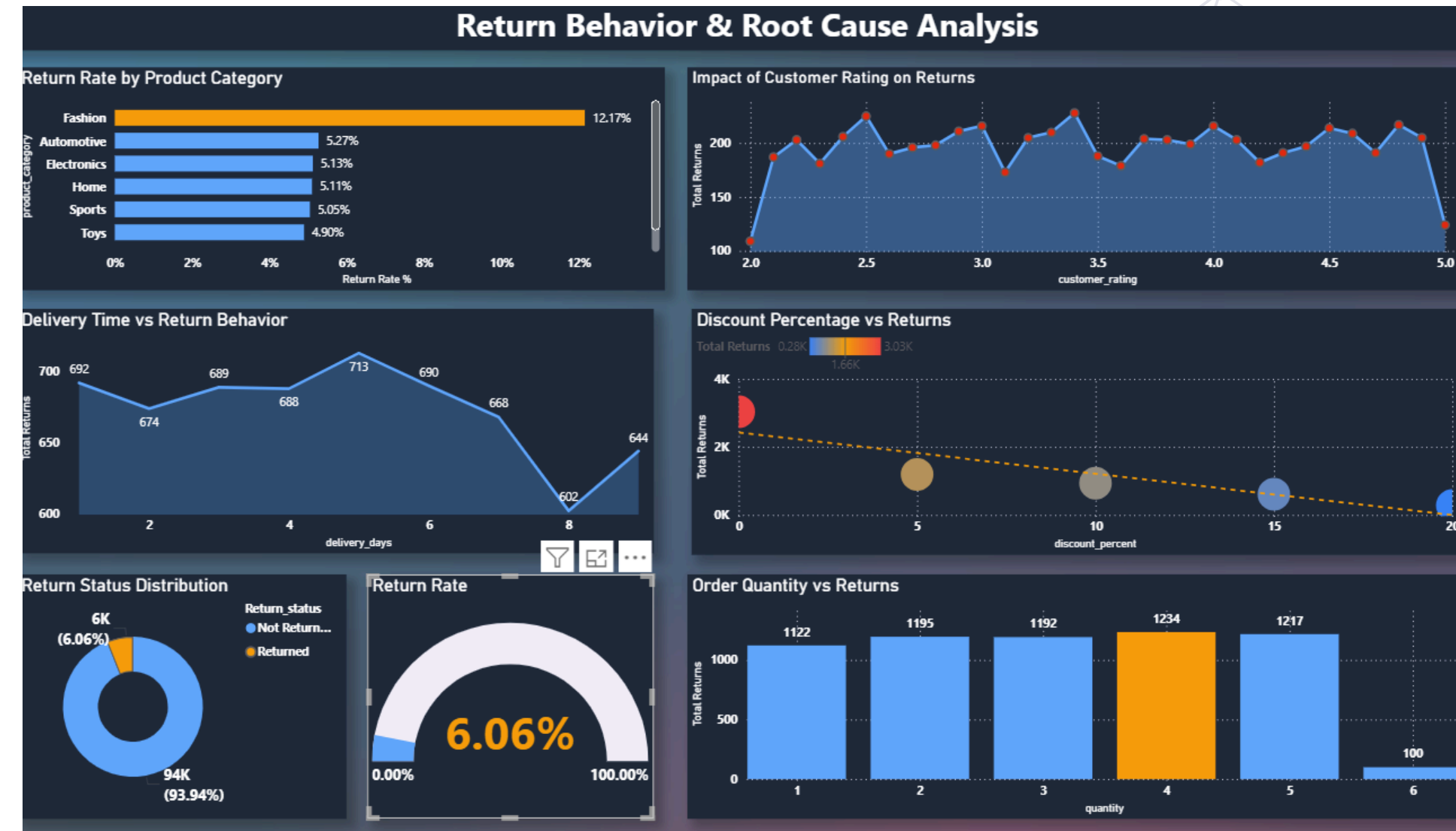
# DASHBOARD 1: E-COMMERCE PERFORMANCE RETURN OVERVIEW

- Provides a high-level overview of business performance
- Displays key KPIs such as Total Revenue, Total Orders, Return Rate, and Return Loss
- Shows return distribution by Product Category and Region
- Analyzes the impact of Discount and Delivery Time on returns
- Displays monthly trends in returns and revenue



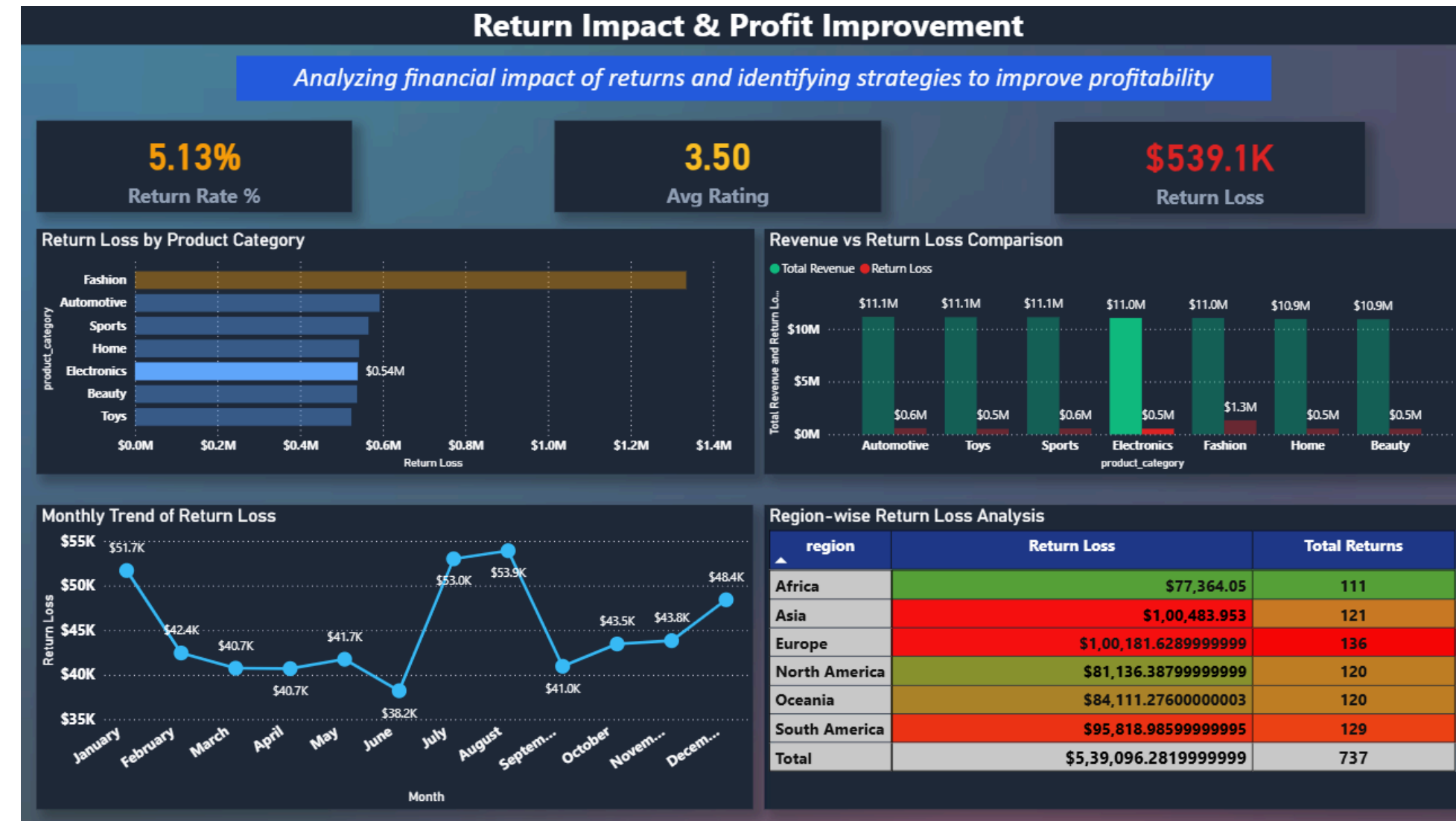
# Dashboard 2: Return Behavior & Root Cause Analysis

- Focuses on identifying factors influencing product returns .
- Analyzes relationship between Customer Rating and Returns .
- Examines impact of Delivery Time on return behavior .
- Evaluates effect of Discount Percentage on returns .
- Studies Order Quantity patterns and return trends



# Dashboard 3: Return Impact & Profit Improvement

- Focuses on financial impact of product returns .
- Displays Return Loss by Product Category .
- Compares Revenue vs Return Loss .
- Analyzes Region-wise return loss .
- Shows Monthly trend of return loss .



# BUSINESS INSIGHTS

- **Fashion category has the highest return rate and return loss .**
- **Delivery delays increase the probability of returns .**
- **Higher discounts are associated with increased return behavior .**
- **Products with lower customer ratings are returned more frequently .**
- **Return patterns vary across regions and months**



# RECOMMENDATIONS

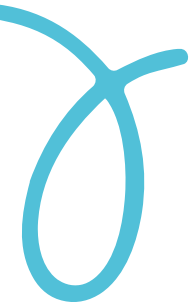
- **Improve product quality in high-return categories (e.g., Fashion) .**
- **Optimize delivery time to reduce delays .**
- **Avoid excessive discount strategies .**
- **Focus on improving customer ratings and feedback .**
- **Strengthen logistics in high-return regions**



# CONCLUSION

- **Product returns have a significant impact on business profitability .**
- **Key factors such as category, delivery time, discount, and rating influence returns .**
- **Identifying these factors helps reduce return rates .**
- **Data-driven decisions can improve customer satisfaction and overall business performance .**





**THANK YOU**